ΔContext: Delayed-Commitment Online Topic Segmentation for Topic-Indexed Conversational Memory

Pravin Raj and Narendra R

***Abstract*—Chat assistants that persist across sessions accumulate histories far longer than their context windows, so every query requires choosing which past turns the model sees. Storing the history as topic segments makes earlier topics cheap to revisit and search, but offline segmenters redraw boundaries as the chat grows, and online rules cut at the first semantic jump. This paper proposes ΔContext, an online segmenter that delays each boundary decision by a few turns, compares the open topic with the turns that follow, and commits a boundary only at a local maximum of semantic distance, never revising it. The segments index a topic-first store. On LongMemEval-S histories of about 490 turns, ΔContext is the most accurate segmenter compared (Pk 0.058 against 0.069 for the best offline method) with zero boundary revisions, against about 17 per 100 turns for offline methods. Its store retrieves as well as a true-session store, and its answers are statistically tied with a full scan's while searching 17% of the turns.**

***Index Terms*—Conversational memory, dialogue topic segmentation, online segmentation, retrieval-augmented generation, large language models, context management.**

I. Introduction

DIALOGUE systems built on large language models (LLMs) are moving from single-session tools to long-lived assistants [1], [2]. An assistant that remembers earlier sessions must manage a memory that grows without bound, and memory has become a central design question for LLM-based agents [3]. Simply enlarging the prompt does not solve it: attention cost grows with input length [4], models use information in the middle of long inputs unreliably [5], and irrelevant or missing context is a known source of hallucination [6].

Practical systems therefore select, for each query, a small set of past turns under a token budget. Retrieval-augmented generation (RAG) embeds stored text and fetches the passages most similar to the query [7], [8], usually from a vector index [9]; summarisation compresses older turns into a running memory [10]; and memory architectures such as MemoryBank and MemGPT combine storage with update and paging rules [11], [12]. The LongMemEval benchmark showed that long-context LLMs lose about 30% accuracy on chat histories of ∼115K tokens and identified the granularity at which history is stored as a key memory design choice [13].

A natural unit of storage for a conversation is the *topic*. Topic-aware modelling improves dialogue comprehension and response selection [14] and response generation [15], dialogue topic segmentation remains an active research problem [16], and retrieval quality depends on how text is chunked before indexing [17]. If a chat history is stored as topic segments as it arrives, an assistant can first find the right topic and then search only inside it, which is cheaper than scanning every turn, and returning to an earlier topic means reopening one stored unit rather than re-reading the history.

Problem statement. Building such a store requires a segmenter that works on a stream. The most accurate unsupervised segmenters are offline: they score every gap of a finished dialogue and then choose boundaries. On a growing chat they must be re-run, and each re-run can move or delete boundaries that already index the store. The simplest online rule, cutting where two consecutive turns differ, decides from a single pair of turns and is noisy. The question this paper addresses is whether an online segmenter that never revises a boundary can match offline segmenters, and whether a store indexed by it retrieves as well as one built from the true topic structure.

Objective. To design an online, training-free topic segmenter whose boundaries are final when made, and to use it as the index of a topic-first conversational memory that keeps retrieval and answer quality at a fraction of the search cost of a full scan.

Contributions.

1) ΔContext, a *delayed-commitment* online segmenter. Each gap between turns is scored after L further turns against the recent part of the open topic, and a boundary is committed only at a local maximum above a threshold. Boundaries are never revised, the decision delay is a constant $L{+}1$ turns, and no training is needed.
2) A topic-first conversational store indexed by these segments, with two-stage retrieval (topic centroids, then turns) under a token budget.
3) A controlled evaluation on four datasets with dev-only tuning, a single test pass and Holm-corrected paired bootstrap tests. It includes an equal-selection protocol, streaming stability, retrieval, scaling to ∼20K turns and end-to-end question answering.
4) Evidence that ΔContext is the most accurate segmenter on long chat histories, including against offline TextTiling (Pk 0.058 vs. 0.069, $p < 10^{-4}$), and the best online method on all four datasets; that it makes zero boundary revisions; and that its store matches a true-session store while answering as well as a full scan from 17.1% of the comparisons.

P. Raj and N. R are with the School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, India (e-mail: pravinraj2054@gmail.com).

Code, configurations and per-question outputs: [repository URL].

Organisation. Section II reviews related work and states the identified gap. Section III presents the four phases of ΔContext. Section IV gives the experimental setup, datasets and results. Section V compares ΔContext with existing works on several metrics, and Section VI concludes.

II. Literature Review

A. Memory for long-running conversations

MemoryBank [11] stores memories with summaries and an Ebbinghaus-style forgetting rule; MemGPT [12] lets the LLM page information between the prompt and external storage; Wang et al. [10] recursively rewrite a summary memory; and LongMemEval [13] proposes session decomposition, fact-augmented keys and time-aware query expansion. None derives topic units from the stream: they store fixed units (turns, rounds, sessions) or LLM-written summaries, which cost an LLM call per update.

B. Dialogue topic segmentation

TextTiling [18] places boundaries at depth-score minima of a similarity curve over a complete text and is still the decoding step of neural methods such as CSM [19], which trains a BERT coherence scorer on automatically built utterance pairs. DialSTART [20] learns topic-aware utterance representations by contrastive learning, and UMLF [16] jointly learns discourse parsing and topic segmentation without labels, reporting the best unsupervised results on DialSeg711 and TIAGE. SuperDialseg [21] shows that supervised models are strong in domain, Ma et al. [14] use unsupervised topic blocks for comprehension, and Nouar and Belhadef [22] audit why a supervised segmenter places its boundaries. All of these segment a complete, short dialogue offline; none addresses a stream whose boundaries index a store.

C. Chunking and label-free semantic structure

Semantic chunking [23] splits text where consecutive embeddings differ by more than a percentile of all distances, so it needs the whole text. Lavarec and Du [17] compare fixed-size, recursive, semantic and LLM-based chunking for RAG and find structure-preserving recursive chunking best, but a chat log lacks the headings and paragraphs it relies on. More broadly, label-free use of distributional and contextual semantics has proven effective for building structure from text: weighted distributional models for named-entity recognition [24], bootstrapped syntax–semantics relation extraction for knowledge-graph construction [25], BERT-strengthened guided topic models for aspect extraction [26], and topic- and concept-guided summarisation with transformers [27]. ΔContext applies the same label-free principle to the geometry of a conversation's turn embeddings. Table I summarises ten recent works most similar to ours.

D. Identified problem

The review exposes four gaps. (i) Conversational memories store fixed units or summaries, not topics found online. (ii) Accurate dialogue segmenters are offline: on a growing chat they must be re-run, which revises boundaries that already index stored data. (iii) Online rules decide from a single pair of turns. (iv) No prior work measures whether online topic segmentation is good enough to index a conversational memory, at scale and end to end. ΔContext targets this combination: online, revision-free, training-free segmentation evaluated as the index of a memory.

III. Proposed Work

A. Overview

ΔContext runs in four phases (Fig. 1). Phases 1–3 run once per incoming turn and maintain the store; Phase 4 runs once per query. A conversation is a stream of turns $u_1, u_2, \ldots$; turn t has a unit-norm embedding $e_t \in \mathbb{R}^d$. Gap g lies between turns g and $g{+}1$, and a boundary at gap g starts a new topic at turn $g{+}1$. The open topic starts at turn a.

B. Phase 1: turn encoding

Each turn is prefixed with its speaker role and encoded by a sentence encoder f (all-MiniLM-L6-v2, d=384 [28]), then normalised: $e_t = f(u_t)/\|f(u_t)\|$. A turn is encoded exactly once, and the same vector serves segmentation (Phase 2) and retrieval (Phase 4).

C. Phase 2: delayed-commitment boundary detection

Let $\mu(i,j)$ be the mean direction of turns $i..j$,

$$\mu(i,j) = \frac{\sum_{k=i}^{j} e_k}{\left\|\sum_{k=i}^{j} e_k\right\|}. \tag{1}$$

The score of gap g compares the recent part of the open topic with the next L turns:

$$s_g = 1 - \mu\big(\max(a,\, g-c+1),\, g\big)^{\top} \mu\big(g+1,\, g+L\big), \tag{2}$$

where c is the left context (the whole open topic if c is unset). After turn t arrives, gap $t{-}L$ gets its score, and gap $g = t{-}L{-}1$, whose neighbours' scores are now both known, is decided. Gap g becomes a boundary if and only if

$$g - a + 1 \geq m, \quad s_g \geq s_{g-1}, \quad s_g \geq s_{g+1}, \quad \mathrm{pass}(s_g), \tag{3}$$

where m is the minimum topic length and pass is either an absolute threshold $s_g \geq \tau$ or a self-calibrating one, $s_g \geq \bar{s}{+}z\cdot \max(\sigma_s, 0.05)$, with $\bar{s}$ and σ_s the mean and standard deviation of the conversation's other gap scores so far (the floor of 0.05 prevents a run of near-identical turns from shrinking the cut-off to nothing). When a boundary is committed, $a \leftarrow g{+}1$, and the scores of later gaps are discarded because they were computed against the old topic. Algorithm 1 gives the full procedure.

Why delay. A rule that cuts at the first large jump cannot tell a topic change from a one-turn digression or a short reply. Waiting L turns compares the topic with what actually follows, and the local-maximum test places the boundary at the sharpest point of the change. **Why a left context.** The centroid of a long topic drifts and becomes insensitive. Comparing against its last c turns tracks the current sub-topic.

TABLE I
TEN RECENT RELATED WORKS: METHOD, ADVANTAGE AND DRAWBACK WITH RESPECT TO AN ONLINE TOPIC-INDEXED CONVERSATIONAL MEMORY.

Work	Year	Algorithm	Advantage	Drawback for our setting
MemoryBank [11]	2024	Memory store with summaries, Ebbinghaus-style forgetting, dense retrieval	Persistent, user-adaptive memory	Flat store of fixed memory units; no topic structure
MemGPT [12]	2023	LLM-managed paging between prompt and external storage	Unbounded virtual context	Every memory operation is an LLM call; no topic structure
Recursive summarisation [10]	2025	LLM rewrites memory from previous memory and new turns	Compact memory, complements long-context LLMs	Lossy; errors accumulate; verbatim evidence is lost
LongMemEval [13]	2025	Benchmark; session decomposition, key expansion, time-aware queries	Systematic memory design study on 115K-token histories	Relies on given session boundaries; flat index over all rounds
DialSTART [20]	2023	Topic-aware contrastive learning of utterance representations	Strong unsupervised accuracy	Needs in-domain training; offline, boundaries change on re-run
SuperDialseg [21]	2023	9,478 labelled dialogues; 18-model benchmark	Supervised models strong in domain	Needs labels; short dialogues; no streaming setting
Topic-aware comprehension [14]	2024	Unsupervised topic blocks + dual-attention matching (TADAM)	Segments aid comprehension and response selection	Segments a finished passage; no store or retrieval cost
UMLF [16]	2025	Mutual learning of discourse parsing and topic segmentation	Best unsupervised Pk on DialSeg711 and TIAGE	Several large encoders; offline decoding
Chunking for RAG [17]	2026	Fixed, recursive, semantic and LLM chunking compared for RAG	Shows chunking choice changes retrieval and answers	Static documents; recursive splitting needs document structure
Boundary explainability [22]	2026	Perturbation-based explanation of a supervised hierarchical segmenter	Audits why boundaries are predicted	Supervised, offline; does not improve segmentation

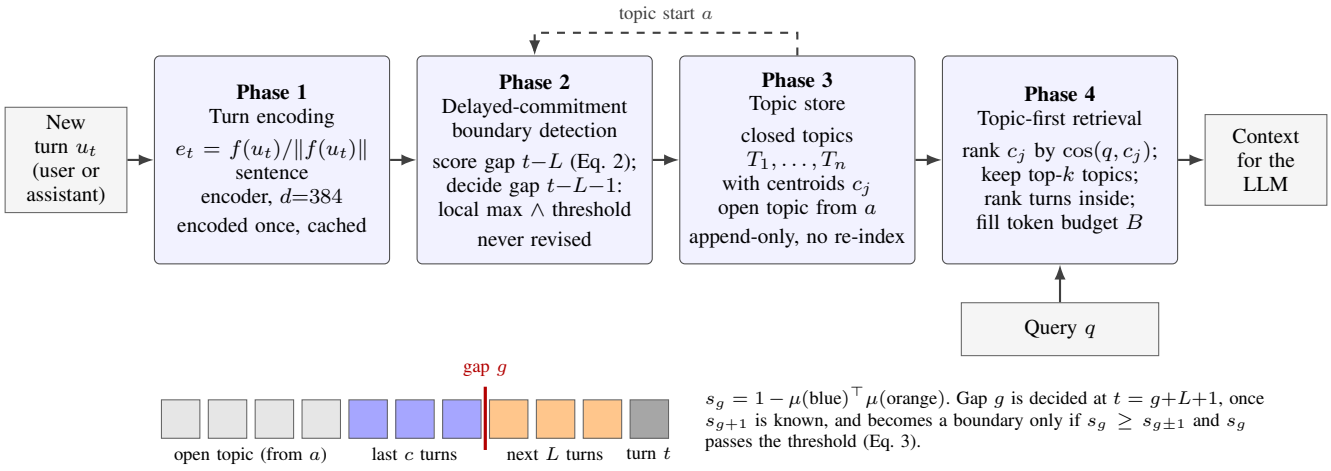


Fig. 1. Architecture of ΔContext: four phases, with Phases 1–3 run per turn and Phase 4 per query. The strip below illustrates delayed commitment with c=L=3: a gap is scored when L later turns exist and decided one turn later, after which the decision is final.

Properties. (P1) *No revision:* the decision on gap g uses only turns up to t and is final; a later commit only moves a forward. (P2) *Constant delay* of L+1 turns. (P3) *Cost* per turn of $O((c+L)\,d)$, independent of history length. (P4) *Training-free*, with four parameters (L, c, threshold, m).

D. Phase 3: topic store

Each closed topic T_j keeps its turn span, its turn embeddings and its centroid $c_j = \mu(T_j)$; the open topic grows turn by turn. Because boundaries are final (P1), the store is append-only. A closed topic is never split, merged or re-indexed, which is what makes the segments usable as a persistent index.

E. Phase 4: topic-first retrieval and context assembly

For a query q (encoded by the same f), Stage 1 ranks topics by $\cos(q, c_j)$ and keeps the top k. Stage 2 ranks the turns inside those topics by $\cos(q, e_i)$. Turns are added to the context in that order until the token budget B is full; a turn that does not fit is skipped. If budget remains, other turns follow by similarity. The selected turns are shown to the LLM in chronological order with their session dates. A query costs $n + \sum_{j \in \text{top-}k} |T_j|$ similarity computations instead of the N of a full scan. This resembles cluster-based (IVF) vector indexes [9], except that the clusters are conversation topics found online, and returning to an old topic means retrieving one stored unit. ΔContext is implemented as a context policy in ContextBench, a harness in which only the policy varies: every method shares the token-budget accounting, the prompt, the encoder and the generator.

Algorithm 1 ΔContext delayed-commitment segmentation (per turn)

Input: embedding e_t of new turn t; parameters L, c, m, threshold rule
Output: committed boundaries $\mathcal{B}$ (never revised)

store e_t; add e_t to the open topic's running sum
$g' \leftarrow t - L$ ▷ newest gap with a full right window
if $g' \geq a$ **then**
  $S[g'] \leftarrow s_{g'}$ by Eq. (2); append $S[g']$ to H
$g \leftarrow g' - 1$ ▷ gap decided now
if $g < a + m - 1$ **or** $g \notin S$ **then**
  return
$p \leftarrow \text{pass}(S[g])$ ▷ τ, or z-score over $H \setminus \{S[g]\}$
if p **and** $S[g] \geq S[g-1]$ **and** $S[g] \geq S[g+1]$ **then**
  $\mathcal{B} \leftarrow \mathcal{B} \cup \{g\}$; close topic $[a, g]$ with centroid $\mu(a, g)$
  $a \leftarrow g + 1$; delete $S[x]$ for all $x \geq a$

IV. Results

A. Experimental setup

Experiments ran on one laptop with an RTX 4060 (8 GB) GPU under Windows 11. Segmentation and retrieval run on the CPU. Answers are generated by Qwen2.5-3B-Instruct [29] (bf16, greedy decoding) with a 2,048-token context budget and the question date in the prompt. Every method's parameters are searched on a development (dev) split only, maximising $J = \frac{1}{2}[(1 - \text{Pk}) + F1_0]$, which guards against the degenerate solutions each metric alone rewards. The chosen setting is then scored once on the test split. For LongMemEval-S, ΔContext uses L=3, c=3, τ=0.75 and m=1.

Metrics. Segmentation: Pk [30] and WindowDiff (WD) [31] with k half the mean reference segment length (lower is better), and boundary F1 with one-to-one matching at tolerance 0 ($F1_0$) and ±1 turn ($F1_1$). Store: topic hit@k (an evidence turn lies in one of the k best topics), topic recall (share of evidence turns that are both in the context and inside the searched topics) and the share of turns searched. Answers: token F1 and exact match (EM) on answerable questions, and the abstention rate on unanswerable ones. Significance: paired bootstrap over conversations or questions (10,000 resamples), Holm-corrected [32] within each dataset or table.

B. Datasets

Table II lists the four datasets. In LongMemEval-S [13] each of 500 questions comes with its own chat history (user–assistant sessions); reference topic boundaries are session boundaries, and evidence turns are labelled. Histories are split by a seeded random permutation into dev and test halves; 100 dev histories are used for tuning. SuperDialseg [21], TIAGE [33] and DialSeg711 [34] are dialogue topic-segmentation benchmarks with annotated boundaries. DialSeg711 has no dev split, so every method uses its SuperDialseg setting there: the dataset is unseen by all methods.

C. Segmentation accuracy

Table III reports ΔContext on all four test sets in three variants: tuned per dataset; one setting for all datasets (the configuration with the best mean dev J over the three dev splits: L=2, c=3, z-score threshold z=1.0); and the previous immediate-decision version (v2). On LongMemEval-S, ΔContext reaches an exact-turn boundary F1 of 0.914 with Pk 0.058. The dialogue benchmarks are much harder for every method because topics last only 3–6 turns. The single setting costs 0.051 J on LongMemEval-S, but it transfers best to the unseen DialSeg711 (Pk 0.283 vs. 0.490 for the SuperDialseg setting), plausibly because the self-calibrating threshold adapts to each conversation's score distribution. Delayed commitment improves on v2 on three of four datasets and ties on TIAGE.

TABLE II
Datasets (test splits). Turns and boundaries are means per conversation.

Dataset	Test	Turns	Bound.	Topic len.	Dev split
LongMemEval-S	250	492.3	46.6	10.3	100 histories
SuperDialseg	1,310	13.1	3.09	3.2	500 (valid.)
TIAGE	100	15.6	3.15	3.8	100 (valid.)
DialSeg711	711	27.2	3.87	5.6	none (unseen)

TABLE III
ΔContext segmentation on the test sets (↓ lower and ↑ higher is better).

Dataset	Variant	Pk↓	WD↓	$F1_0$ ↑	$F1_1$ ↑	J ↑
LongMemEval-S	v3 tuned	**0.058**	**0.077**	**0.914**	**0.917**	**0.928**
	v3 one setting	0.094	0.139	0.848	0.852	0.877
	v2 (no delay)	0.102	0.133	0.825	0.838	0.861
SuperDialseg	v3 tuned	0.413	0.421	**0.502**	**0.689**	**0.544**
	v3 one setting	**0.408**	**0.409**	0.248	0.417	0.420
	v2 (no delay)	0.459	0.489	0.437	0.666	0.489
TIAGE	v3 tuned	0.427	0.456	0.365	0.570	**0.469**
	v3 one setting	**0.388**	**0.400**	0.290	0.443	0.451
	v2 (no delay)	0.454	0.496	**0.385**	**0.605**	0.466
DialSeg711	v3 (SD setting)	0.490	0.619	0.497	0.560	0.503
	v3 one setting	**0.283**	**0.309**	**0.554**	**0.626**	**0.636**
	v2 (SD setting)	0.522	0.695	0.455	0.509	0.466

D. Ablation: which component matters

Table IV swaps one component at a time from the chosen configuration, re-tunes the threshold on dev, and scores once on test. Delayed commitment is the decisive component. Replacing it with v2's immediate decision costs 0.066 J on LongMemEval-S and 0.056 on SuperDialseg, and a look-ahead that does not match the topic length costs as much (on LongMemEval-S L=1 loses 0.051; on SuperDialseg, with 3-turn topics, L=2 or 3 loses 0.06). The other two choices matter less. The left context changes J by at most 0.016, and the z-score threshold costs 0.014–0.016 when tuning per dataset, yet it is what makes the single setting transfer.

E. Retrieval from the topic store and question answering

Table V evaluates the store on the 470 answerable LongMemEval-S questions with top-3 topics and a 2,048-token budget. The ΔContext store puts an evidence turn in the first-ranked topic for 77.0% of questions and in the top three for 93.0%, while computing 17.1% of a full scan's

TABLE IV
ONE-COMPONENT ABLATION (TEST J; Δ = VARIANT − CHOSEN, 95% BOOTSTRAP CI; ALL $p < 10^{-3}$).

LongMemEval-S			SuperDialseg		
Variant	J	Δ	Variant	J	Δ
chosen (L=3)	0.928	–	chosen (L=1)	0.544	–
$L = 1$	0.877	−0.051	$L = 2$	0.480	−0.065
$L = 2$	0.921	−0.007	$L = 3$	0.483	−0.061
whole-topic left	0.912	−0.016	last-3 left	0.553	+0.008
z-score threshold	0.912	−0.016	z-score threshold	0.530	−0.014
no delay (v2)	0.861	−0.066	no delay (v2)	0.489	−0.056

TABLE V
TOPIC STORE (TOP-3) AND QA ON LONGMEMEVAL-S (470 ANSWERABLE QUESTIONS, BUDGET 2,048). $^{***}p < 10^{-3}$ VS. THE ΔCONTEXT STORE.

Store	Hit@1	Hit@3	Rec.	Search.	F1	EM
ΔContext v3 (ours)	0.770	**0.930**	0.732	17.1%	**0.302**	0.206
Session oracle	**0.785**	0.926	0.733	16.5%	0.298	0.202
v2 detector	0.749	0.917	0.716	–	–	–
v2 w/o localisation	0.426***	0.732***	0.516***	–	–	–
v1 path-length	0.640***	0.847***	0.609***	–	–	–
Full scan (all turns)	–	–	**0.774*****	100%	0.311	**0.217**

similarities. A store built from the true session boundaries (a reference, not a strict bound) reaches 78.5% and 92.6% (differences not significant), and topic recall is the same (0.732 vs. 0.733). Boundary placement is what drives this: the earlier v2 and path-length detectors, and v2 without boundary localisation, lose up to 0.34 hit@1. Scanning every turn keeps more evidence (0.774 vs. 0.732, $p < 10^{-3}$), but the answers do not differ significantly. ΔContext's F1 is 0.302 against 0.311 for the full scan, a difference of -0.009 (95% CI $[-0.028, +0.011]$), and the store abstains on all 30 unanswerable questions (full scan: 28). Giving the reader only the evidence sessions (∼6K tokens) yields 0.279, so with a 3B reader the answer stage, not retrieval, limits accuracy. Because segmentation parameters were tuned on dev histories, we re-scored the same outputs on the 237 test-half questions only, and the conclusions are unchanged: hit@1 0.776 (oracle 0.793, n.s.) and F1 0.331 vs. 0.337 for the full scan (n.s.).

F. Scaling the store

To test a store far larger than one history, each history was merged with up to 39 other histories as distractors, giving 492 to 19,732 turns (Fig. 2; 237 test-half questions). ΔContext tracks the session-oracle store at every size (hit@3 0.924, 0.827, 0.692, 0.612, 0.481 against 0.924, 0.844, 0.713, 0.616, 0.477). Its query cost falls from 17.1% of a full scan at 492 turns to 9.9% at 19.7K turns (1,949 of 19,732 comparisons), because topics grow in number but each query reads only three of them.

V. COMPARISONS WITH EXISTING WORKS

A. Compared works and metrics

ΔContext is compared with three existing families, all run on the same MiniLM embeddings, the same splits and the same tuning protocol: (a) **TextTiling** [18], the decoding step

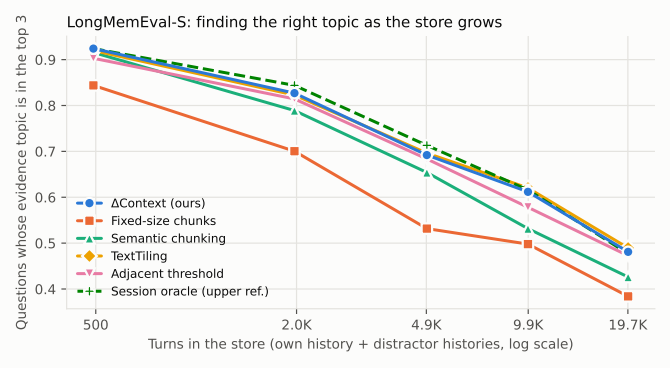


Fig. 2. Topic hit@3 as the store grows with distractor histories (LongMemEval-S, 237 test-half questions). ΔContext stays on the session-oracle curve; fixed-size chunks and semantic chunking fall behind.

of CSM and UMLF [16], [19] and the strongest unsupervised offline baseline available on the same encoder; (b) **semantic chunking** [23], the percentile-threshold splitter evaluated for RAG by Lavarec and Du [17]; and (c) **full-scan turn-level dense retrieval**, the round-level index of LongMemEval [13] and the retrieval mode of MemoryBank [11]. Two common online rules (adjacent-turn threshold, fixed-size chunks) complete the field, and published trained segmenters are compared in Section V-F. Metrics cover segmentation (Pk, WD, $F1$, J), streaming (revisions, delay), retrieval (hit@k, topic recall, search share) and answers (F1, EM).

B. Segmentation, each method tuned per dataset

Table VI and Fig. 3(a) compare all methods. On LongMemEval-S, ΔContext is the most accurate on every metric. Against TextTiling, Pk is lower by 0.011 (95% CI $[-0.014, -0.007]$) and $F1_0$ higher by 0.020 ($[+0.015, +0.024]$), both $p < 10^{-4}$. Against semantic chunking and the adjacent threshold, Pk is lower by 0.083 and 0.064 and $F1_0$ higher by 0.158 and 0.160. It is also the best online method on all four datasets. On SuperDialseg, offline TextTiling has a lower Pk (by 0.028, $p < 10^{-4}$), while $F1_0$ does not differ (+0.008, n.s.). On TIAGE, ΔContext, TextTiling and semantic chunking are statistically tied. On the unseen DialSeg711, TextTiling's SuperDialseg setting transfers better than ΔContext's.

C. Segmentation under equal selection

A deployed assistant cannot be re-tuned for each user, so we also give every method a single setting chosen by the same rule (best mean dev J over the three dev splits). Table VII and Fig. 3(b) show the result. ΔContext beats every baseline on LongMemEval-S by at least 0.080 J ($p < 10^{-4}$). It also beats semantic chunking, the adjacent threshold and fixed chunks on the unseen DialSeg711, where it ties TextTiling (-0.008, n.s.; ΔContext has the lower Pk, 0.283 vs. 0.295). On TIAGE all methods except fixed chunks tie. On SuperDialseg, whose topics are about three turns long, ΔContext's single setting (L=2) trails TextTiling and fixed chunks. Notably, TextTiling with one setting falls from 0.913 to 0.790 J on

TABLE VI
TOPIC SEGMENTATION ON FOUR TEST SETS, EACH METHOD TUNED ON THAT DATASET'S DEV SPLIT (DIALSEG711: SUPERDIALSEG SETTING). CELLS: PK↓ / $F1_0$ ↑ (J ↑). BOLD: BEST J PER DATASET. "ONLINE": DECIDES EACH BOUNDARY WITHOUT LATER TURNS BEING ABLE TO CHANGE IT.

Method	Online	Training	LongMemEval-S	SuperDialseg	TIAGE	DialSeg711
ΔContext v3 (ours)	yes	none	**0.058 / 0.914 (0.928)**	0.413 / 0.502 (0.544)	0.427 / 0.365 (0.469)	0.490 / 0.497 (0.503)
TextTiling [18]	no	none	0.069 / 0.894 (0.913)	**0.385 / 0.493 (0.554)**	0.442 / 0.374 (0.466)	**0.401 / 0.560 (0.579)**
Semantic chunking [23]	no	none	0.141 / 0.756 (0.808)	0.498 / 0.399 (0.450)	**0.441 / 0.397 (0.478)**	0.464 / 0.448 (0.492)
Adjacent-turn threshold	yes	none	0.122 / 0.754 (0.816)	0.511 / 0.408 (0.449)	0.471 / 0.399 (0.464)	0.499 / 0.417 (0.459)
Fixed-size chunks	yes	none	0.532 / 0.170 (0.319)	0.495 / 0.455 (0.480)	0.544 / 0.260 (0.358)	0.576 / 0.457 (0.441)

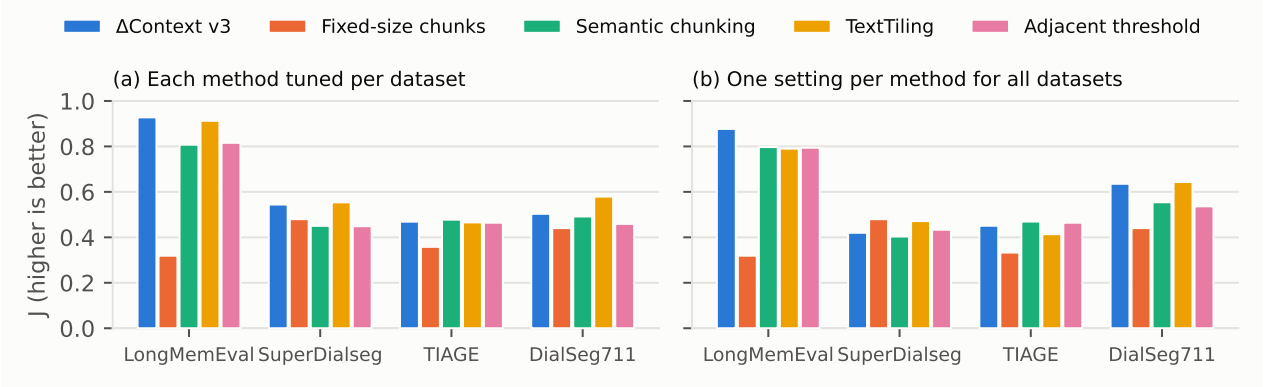


Fig. 3. Segmentation score J on four test sets. (a) Each method tuned per dataset. (b) One setting per method, chosen by mean dev J over LongMemEval-S, SuperDialseg and TIAGE and used unchanged everywhere. Significance tests for (b) are in Table VII.

TABLE VII
EQUAL SELECTION: ΔJ (ΔCONTEXT − BASELINE) ON TEST, ONE SETTING PER METHOD. *HOLM $p < 0.05$; OTHERWISE N.S. W/T/L: SIGNIFICANT WINS/TIES/LOSSES.

Baseline	LongMemEval	SuperDial.	TIAGE	DialSeg711	W/T/L
TextTiling	$+0.087^*$	-0.052^*	$+0.037$	-0.008	1/2/1
Semantic chunk.	$+0.080^*$	$+0.016$	-0.018	$+0.081^*$	2/2/0
Adjacent thr.	$+0.083^*$	-0.014	-0.013	$+0.099^*$	2/2/0
Fixed chunks	$+0.558^*$	-0.060^*	$+0.118^*$	$+0.195^*$	3/0/1

LongMemEval-S, whereas ΔContext falls only from 0.928 to 0.877, so ΔContext is the more robust choice when one configuration must serve both long chats and short dialogues.

D. Streaming stability

We replayed the 250 LongMemEval-S test histories one turn at a time, re-running the offline methods after every turn (Table VIII, "Revisions" and "Delay"). TextTiling revises 17.0 existing boundaries per 100 new turns (14.0 removed, 3.0 late cuts) and semantic chunking 17.6. Each revision re-indexes stored topics and invalidates any summary or cache attached to them. ΔContext revises none, by construction (P1), and still ends with the highest boundary F1 (0.917 vs. 0.899). Its median detection delay is 4 turns (L+1), against 5 for TextTiling, which needs a full block of turns on both sides of a gap.

E. Retrieval cost and answer quality

Table VIII and Fig. 4 compare the memories built on each segmenter with full-scan retrieval. ΔContext and TextTiling

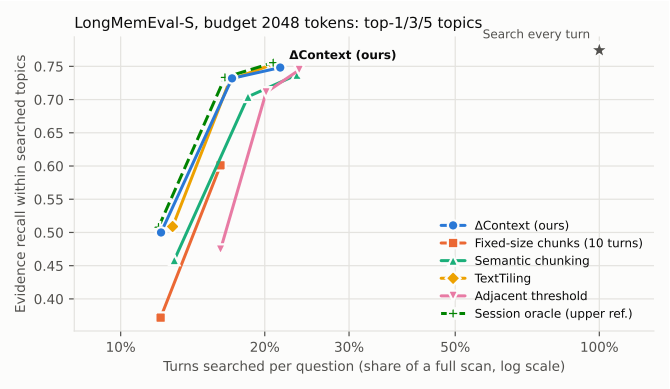


Fig. 4. Evidence recall against the share of turns searched (top-1/3/5 topics; LongMemEval-S, 2,048-token budget). The ΔContext store reaches the session-oracle trade-off; a full scan reads every turn.

give statistically indistinguishable stores, but only ΔContext's is built without revisions. ΔContext leads semantic chunking on hit@1 (+0.049; not significant after Holm correction, $p = 0.013$ uncorrected on the test half) at a lower search share, and beats 10-turn chunks by 0.23 hit@1 and 0.04 F1. The full scan keeps the most evidence but reads every turn, and its answer F1 is within the confidence interval of ΔContext's. In Fig. 4, the ΔContext curve lies on the session-oracle curve at every top-k. At 19.7K turns (Fig. 2), ΔContext also leads semantic chunking (hit@3 0.481 vs. 0.426) and fixed chunks (0.384), and matches TextTiling (0.489), which performs 10% more comparisons.

TABLE VIII
MEMORY COMPARISON ON LONGMEMEVAL-S (BUDGET 2,048, TOP-3 TOPICS, 470 ANSWERABLE QUESTIONS). PK: FINAL SEGMENTATION IN THE STREAMING REPLAY. "REVISIONS": BOUNDARIES CHANGED PER 100 NEW TURNS. $^{***}p < 10^{-3}$, $^{*}p < 0.05$ VS. ΔCONTEXT.

Memory	Online	Revisions	Delay	Pk	Hit@1	Hit@3	Topic rec.	Searched	QA F1	QA EM
ΔContext v3 store (ours)	yes	**0**	4	**0.058**	0.770	**0.930**	0.732	17.1%	0.302	0.206
TextTiling store [18]	no	17.0	5	0.069	**0.777**	0.921	0.729	16.9%	0.296	0.200
Semantic chunking store [17], [23]	no	17.6	1†	0.141	0.721	0.904	0.704	18.4%	0.284	0.194
Adjacent-threshold store	yes	0	1	0.122	0.736	0.902	0.711	20.1%	–	–
Fixed-size chunks (10 turns)	yes	0	1	0.500	0.543***	0.813***	0.601***	16.2%	0.263*	0.187
Full scan, turn level [11], [13]	yes	–	–	–	–	–	**0.774*****	100%	**0.311**	**0.217**
Session oracle (reference)	–	–	–	0	0.785	0.926	0.733	16.5%	0.298	0.202

†Median; semantic chunking's mean delay is 14.5 turns because many boundaries appear only after later percentile shifts.

TABLE IX
1−PK / 1−WD (%, HIGHER IS BETTER). PUBLISHED ROWS FROM [16]; DIFFERENT ENCODERS AND TRAINING, SO INDICATIVE ONLY.

Method	Trained / online	TIAGE	DialSeg711
TeT+CLS [34]	no / no	52.7 / 48.6	59.5 / 56.9
CSM [19]	yes / no	52.8 / 49.1	75.7 / 73.7
DialSTART [20]	yes / no	55.7 / 50.2	82.1 / 80.2
UMLF [16]	yes / no	59.3 / 55.6	**85.9 / 84.2**
TextTiling, MiniLM (our run)	no / no	56.1 / 54.8	70.5 / 67.8
ΔContext v3, one setting	no / **yes**	**61.2 / 60.0**	71.7 / 69.1

F. Comparison with published dialogue segmenters

Table IX places ΔContext next to published unsupervised segmenters, as reported by Xu et al. [16]. These methods train neural scorers or encoders on in-domain dialogues and decode offline. ΔContext uses a 22M-parameter off-the-shelf encoder, no training and online decisions. On TIAGE, the single ΔContext setting scores 61.2 (1−Pk), above the 59.3 that UMLF reports. On DialSeg711 it scores 71.7, above TeT+CLS [34] but 4–14 points below CSM, DialSTART and UMLF. The numbers are indicative only: encoders, training data and Pk window conventions differ, and the TIAGE dev split took part in choosing the single setting.

G. Summary of the comparison

On long chat histories, the setting a conversational memory faces, ΔContext beats offline TextTiling under both tuning protocols and every online rule. It is the only accurate method that never revises a boundary, so its segments can serve as a persistent index, and its store matches a true-session store while answering as well as a full scan at about a sixth of the cost. Its limits: on short dialogue benchmarks it ties offline TextTiling (TIAGE, DialSeg711) or trails it (SuperDialseg); trained offline segmenters are more accurate on DialSeg711; and a full scan keeps 4 points more evidence. Delayed commitment was introduced after the immediate-decision v2 lost on the dialogue benchmarks, which is why all comparisons use dev-only tuning and one test pass.

VI. CONCLUSION

This paper presented ΔContext, which stores a growing chat as topic segments found online and retrieves by topic first. Its novelty is delayed commitment: each gap is judged against the turns that follow it and committed only at a local maximum, so boundaries are final, arrive after a constant delay and need no training. On long chat histories ΔContext was the most accurate segmenter compared, including offline TextTiling, while revising no boundary where offline methods revised about seventeen per hundred turns. The resulting store matched a true-session store up to twenty thousand turns and answered as well as scanning every turn while searching about a sixth of them. On short dialogue benchmarks it was the best online method but tied or trailed the best offline one. Future work will adapt the look-ahead to topic length within a conversation, and attach compact per-topic summaries and hierarchical topic links so that very large stores can be searched in one step.

REFERENCES

[1] J. Ni, T. Young, V. Pandelea, F. Xue, and E. Cambria, "Recent advances in deep learning based dialogue systems: A systematic survey," *Artificial Intelligence Review*, vol. 56, no. 4, pp. 3055–3155, 2023, doi: 10.1007/s10462-022-10248-8.

[2] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, and Z. Wei, "A survey on large language model based autonomous agents," *Frontiers of Computer Science*, vol. 18, no. 6, Art. no. 186345, 2024, doi: 10.1007/s11704-024-40231-1.

[3] Z. Zhang, Q. Dai, X. Bo, C. Ma, R. Li, X. Chen, J. Zhu, Z. Dong, and J.-R. Wen, "A survey on the memory mechanism of large language model-based agents," *ACM Transactions on Information Systems*, vol. 43, no. 6, pp. 1–47, 2025, doi: 10.1145/3748302.

[4] Y. Tay, M. Dehghani, D. Bahri, and D. Metzler, "Efficient transformers: A survey," *ACM Computing Surveys*, vol. 55, no. 6, pp. 1–28, 2022, doi: 10.1145/3530811.

[5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024, doi: 10.1162/tacl_a_00638.

[6] L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," *ACM Transactions on Information Systems*, vol. 43, no. 2, pp. 1–55, 2025, doi: 10.1145/3703155.

[7] S. Wu, Y. Xiong, Y. Cui, H. Wu, C. Chen, Y. Yuan, L. Huang, X. Liu, T.-W. Kuo, N. Guan, and C. J. Xue, "Retrieval-augmented generation for natural language processing: A survey," *Artificial Intelligence Review*, vol. 59, no. 9, Art. no. 192, 2026, doi: 10.1007/s10462-026-11605-7.

[8] W. X. Zhao, J. Liu, R. Ren, and J.-R. Wen, "Dense text retrieval based on pretrained language models: A survey," *ACM Transactions on Information Systems*, vol. 42, no. 4, pp. 1–60, 2024, doi: 10.1145/3637870.

[9] J. J. Pan, J. Wang, and G. Li, "Survey of vector database management systems," *The VLDB Journal*, vol. 33, no. 5, pp. 1591–1615, 2024, doi: 10.1007/s00778-024-00864-x.

[10] Q. Wang, Y. Fu, Y. Cao, S. Wang, Z. Tian, and L. Ding, "Recursively summarizing enables long-term dialogue memory in large language models," *Neurocomputing*, vol. 639, Art. no. 130193, 2025, doi: 10.1016/j.neucom.2025.130193.

[11] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing large language models with long-term memory," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 38, no. 17, pp. 19724–19731, 2024, doi: 10.1609/aaai.v38i17.29946.

[12] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," arXiv:2310.08560, 2023.

[13] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2025.

[14] X. Ma, Y. Xu, H. Zhao, and Z. Zhang, "Multi-turn dialogue comprehension from a topic-aware perspective," *Neurocomputing*, vol. 578, Art. no. 127385, 2024, doi: 10.1016/j.neucom.2024.127385.

[15] X. Sun and B. Ding, "Neural network with hierarchical attention mechanism for contextual topic dialogue generation," *IEEE Access*, vol. 10, pp. 4628–4639, 2022, doi: 10.1109/ACCESS.2022.3140820.

[16] J. Xu, F. Jiang, A. Gao, L. F. D'Haro, and H. Li, "Unsupervised mutual learning of discourse parsing and topic segmentation in dialogue," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 33, pp. 4667–4681, 2025, doi: 10.1109/TASLPRO.2025.3623884.

[17] E. Lavarec and Y. Du, "Evaluating document chunking approaches for retrieval augmented generation in editorial content," *IAES International Journal of Artificial Intelligence*, vol. 15, no. 2, pp. 1909–1918, 2026, doi: 10.11591/ijai.v15.i2.pp1909-1918.

[18] M. A. Hearst, "TextTiling: Segmenting text into multi-paragraph subtopic passages," *Computational Linguistics*, vol. 23, no. 1, pp. 33–64, 1997.

[19] L. Xing and G. Carenini, "Improving unsupervised dialogue topic segmentation with utterance-pair coherence scoring," in *Proc. 22nd Annu. Meeting Special Interest Group on Discourse and Dialogue (SIGDIAL)*, 2021, pp. 167–177, doi: 10.18653/v1/2021.sigdial-1.18.

[20] H. Gao, R. Wang, T.-E. Lin, Y. Wu, M. Yang, F. Huang, and Y. Li, "Unsupervised dialogue topic segmentation with topic-aware contrastive learning," in *Proc. 46th Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2023, pp. 2481–2485, doi: 10.1145/3539618.3592081.

[21] J. Jiang, C. Dong, S. Kurohashi, and A. Aizawa, "SuperDialseg: A large-scale dataset for supervised dialogue segmentation," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 4086–4101, doi: 10.18653/v1/2023.emnlp-main.249.

[22] F. Nouar and H. Belhadef, "Boundary-centered explainability for dialogue topic segmentation," *Electronics*, vol. 15, no. 17, Art. no. 3998, 2026, doi: 10.3390/electronics15173998.

[23] LangChain, "SemanticChunker (langchain_experimental.text_splitter)," 2024. [Online]. Available: https://python.langchain.com

[24] G. Veena, V. Kanjirangat, and D. Gupta, "AGRONER: An unsupervised agriculture named entity recognition using weighted distributional semantic model," *Expert Systems with Applications*, vol. 229, Art. no. 120440, 2023, doi: 10.1016/j.eswa.2023.120440.

[25] G. Veena, D. Gupta, and V. Kanjirangat, "Semi-supervised bootstrapped syntax-semantics-based approach for agriculture relation extraction for knowledge graph creation and reasoning," *IEEE Access*, vol. 11, pp. 138375–138398, 2023, doi: 10.1109/ACCESS.2023.3339552.

[26] M. Venugopalan and D. Gupta, "An enhanced guided LDA model augmented with BERT based semantic strength for aspect term extraction in sentiment analysis," *Knowledge-Based Systems*, vol. 246, Art. no. 108668, 2022, doi: 10.1016/j.knosys.2022.108668.

[27] B. M. Gurusamy, P. K. Rengarajan, and P. Srinivasan, "A hybrid approach for text summarization using semantic latent Dirichlet allocation and sentence concept mapping with transformer," *International Journal of Electrical and Computer Engineering*, vol. 13, no. 6, pp. 6663–6672, 2023, doi: 10.11591/ijece.v13i6.pp6663-6672.

[28] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. Conf. Empirical Methods in Natural Language Processing and Int. Joint Conf. Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3980–3990, doi: 10.18653/v1/D19-1410.

[29] Qwen Team, "Qwen2.5 technical report," arXiv:2412.15115, 2024.

[30] D. Beeferman, A. Berger, and J. Lafferty, "Statistical models for text segmentation," *Machine Learning*, vol. 34, pp. 177–210, 1999.

[31] L. Pevzner and M. A. Hearst, "A critique and improvement of an evaluation metric for text segmentation," *Computational Linguistics*, vol. 28, no. 1, pp. 19–36, 2002.

[32] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[33] H. Xie, Z. Liu, C. Xiong, Z. Liu, and A. Copestake, "TIAGE: A benchmark for topic-shift aware dialog modeling," in *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021, pp. 1684–1690, doi: 10.18653/v1/2021.findings-emnlp.145.

[34] Y. Xu, H. Zhao, and Z. Zhang, "Topic-aware multi-turn dialogue modeling," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 35, no. 16, pp. 14176–14184, 2021, doi: 10.1609/aaai.v35i16.17668.